

Transformer Differential Protection with Neural Network Based Inrush Stabilization

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Abstract — Application of artificial neural networks (ANN) for transformer differential protection stabilization against inrush conditions is presented. Three versions of the stabilization scheme are described. The best of them employs three ANNs fed with transformer terminal currents that has proven to be superior over the two other ANN schemes. The final solution combines the classification strengths of neural networks with commonly used second harmonic restraint, thus being a hybrid classification unit. To determine the most suitable ANN topology for the inrush classifier a genetic algorithm was used. The developed optimized neural inrush detection units have been tested with EMTP-ATP generated signals, proving better performance than traditionally used stabilization algorithms.

Index Terms — protective relaying, transformer differential protection, artificial neural networks, genetic algorithms.

I. INTRODUCTION

It has been proved that considerable improvement of operation as well as quite simple achievement of adaptive features of protection functions may be obtained with use of various Artificial Intelligence (AI) techniques [1 – 4]. The most effective and commonly recognised AI methods include Artificial Neural Networks (ANN), Fuzzy Logic (FL) systems and Expert Systems (ES). Mentioned intelligent techniques have gained remarkable attention in the best research centres all around the world for more than 15 years. Noticeable is also the interest of world-renowned technical bodies and organisations such as: CIGRE committees (SC38 Power System Analysis and Techniques, SC39 Power System Operation and Control, SC34 Protection), IEEE and IEEE Power Engineering Society, that have established special working groups and task committees to investigate possibility of AI application for solving problems related to power system planning, development, management and operation. The majority of AI techniques usage in power system protection and control constitute applications of neural networks and expert systems. In this paper application of Artificial Neural Networks to inrush discrimination in power transformers is described.

Although differential protection has been successfully used for decades to protect power transformers against faults, its implementation (even in digital technique) is not free from errors which result in improper functioning either as undesired delay or lack of tripping for internal faults or as unwanted tripping for magnetizing inrush conditions. During normal operation of power transformers the magnetizing currents are very small, usually below 1% of the rated currents. However, due to nonlinear magnetizing characteristic an increase of the core

flux amplitude by 10 % causes an increase of the magnetizing current 3-5 times, while an increase by 20% magnifies the current 10-20 times. In the latter case the harmonic spectrum of the current shows domination of the fundamental harmonic, however the 2nd harmonic exceeds 30-40% of the fundamental one. Magnetising inrush currents caused by high DC components of the flux, which result from sudden increase of the terminal voltages, may also become very high. The current level depend on various factors, however the dominant ones are point on wave of the voltage increase, residual flux in the core as well as source impedances. Excessive magnetizing currents may also arise as a result of voltage recovery after clearing of a nearby fault, change of the character of a fault or out-of-phase synchronizing of a connected generator.

Numerous single or compound criteria are usually used or proposed in the literature to discriminate inrush conditions and to prevent unwanted protection tripping. They are based on:

- second harmonic restraint [5],
- current waveform analysis (flat period observed) [6],
- model based methods [7],
- flux restraint [8], etc.

Since all the methods mentioned display certain limitations and cannot classify all possible inrush cases correctly, new methods for inrush recognition are still being developed. Great hope is connected with application of Artificial Intelligence techniques, i.e. fuzzy logic [9] or, here – neural networks.

This paper starts with description of several versions of the ANN based stabilization units (Section II). Then the scheme testing with numerous EMTP-ATP simulation cases is presented (Section III). In Section IV the ANN structure optimization with proposed genetic procedure is presented, which is followed by testing of the final solution with selected difficult simulation and real-world registered inrush cases (Section V). Section VI closes the paper with some conclusions and final application recommendations.

II. MAGNETIZING INRUSH DETECTION WITH ANNS

Artificial Neural Networks represent a modern approach to problem solving also for power system protection and control applications. The ANNs perform actions similar to human reasoning which rely on experience gathered during so called training. Advantages of neural computing methodologies over conventional approaches include faster computation, learning ability, robustness and noise rejection. Once trained the ANN should possess the feature of knowledge generalisation, which

means that they should reasonably respond to the situations (from certain space) that had not been presented during training. The ANNs are mainly used for classification and/or decision-making in case of problems that are not fully described in the deterministic way or when their description (model) is non-linear or heavy complicated.

The general scheme of the ANN based protection is shown in Fig. 1. The protected transformer is to be switched-off when the relay is activated (block 1 – high differential current detected), external fault is ruled out (block 2 – percentage stabilization exceeded) and inrush conditions are excluded (block 3 – traditionally 2nd harmonic ratio is checked). Additional path for tripping is when the threshold of overcurrent support is exceeded (block 4) which is usually applied to speed-up the relay response for clear internal faults. With such an arrangement the fundamental and 2nd harmonic components of the differential currents (plus fifth for overexcitation checking, not shown in Fig. 1) as well as amplitudes of 50Hz components of the through currents are to be extracted, which is often done with use of full cycle Fourier filters with well-known band-pass frequency responses and limited dynamical properties resulting from their data window lengths [10].

In this paper it is proposed that the inrush stabilisation task is entrusted to an ANN at the output of which a decision as to inrush/non-inrush is issued. Three versions of ANN-based decision supporting units for inrush recognition have been investigated and are described further. The ANN schemes developed were:

- A – single-ANN scheme fed with differential currents,
- B – three-ANN scheme fed with differential currents,
- C – three-ANN scheme fed with terminal currents.

In this study feedforward ANNs of perceptron type were used, whereas their size, i.e. number of layers and neurones in particular layers, has been optimised. The training examples were prepared in simulative way, while for testing purpose both simulation and real-world (registered) cases were used.

A. Single-ANN three-phase scheme

The first and most compact version of neural stabilisation block was composed of a single ANN that was fed with the samples of differential currents from all three phases. The ANN input vector consisted of 30 samples, 10 from each phase, observed within sliding 20ms-long data window. Every second sample was taken only (loosing a part of information on the current waveshapes is not big) to save the processing time and to keep the ANN structure reasonably small. The four-layer ANN with 18-12-6-1 neurones in particular layers was selected at the initial stage of research. The activation function of all the neurones was a bipolar sigmoidal function (Fig. 3). The ANN was trained to produce output equal +1 for inrush cases and -1 for other situations. The decision threshold Δ was set at 0.0, which means that for positive output values an inrush was confirmed and the relay was blocked. The investigations have shown that the single-ANN stabilisation unit performed well for all simulated fault cases (the relay was restrained from tripping), however not all inrush cases were

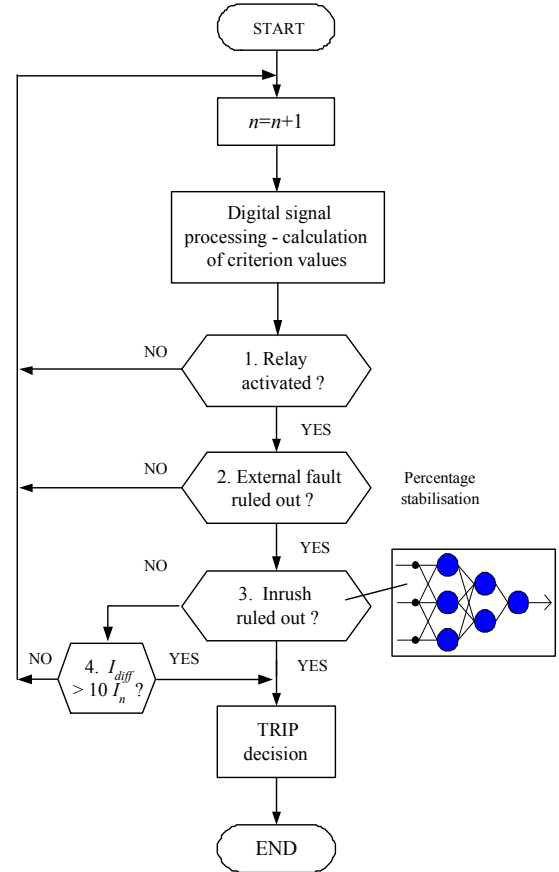


Fig. 1. Block scheme of differential protection with neural stabilization.

properly recognized (overfunction in some cases). Adjusting the threshold value did not bring much improvement, thus another attempt to use three ANNs for phase signals was studied.

B. Three-ANN scheme fed with differential currents

The other neural protection scheme developed consisted of three ANNs, each one fed with the samples of differential current from the respective phase. The outputs of particular ANNs were averaged to form a decision signal. Mean value of the ANN outputs was then compared with the threshold set as for single-ANN scheme. The ANNs used in the scheme were smaller than in single-ANN version, they consisted of 16-8-4-1 neurones in particular layers. Training of the ANNs was done in supervised manner with the current patterns cut from simulated waveforms. The tests revealed that for such a solution all simulated transformer energisation cases were correctly classified. At the same time no deterioration in recognition of fault cases was observed. On the contrary, the fault cases were in most situations confirmed faster than with use of 2nd harmonic restraint that was blocking the relay for longer time. Statistics of the relay operation can be seen in Table I, along with the results for other types of neural schemes developed.

C. Three-ANN single-phase scheme with terminal signals

The block scheme of the neural stabilisation unit with three ANNs fed with transformer terminal currents is shown in Fig. 2. By designing the scheme both definition of the target ANN

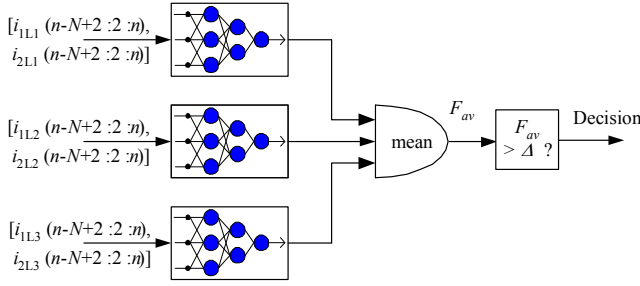


Fig. 2. Three-ANN single-phase solution with transformer terminal currents.

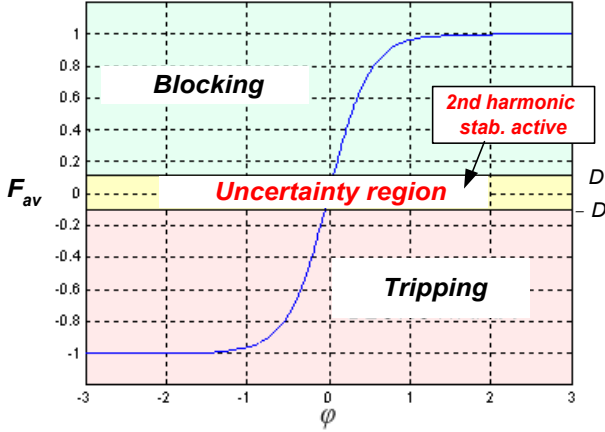


Fig. 3. Illustration of the neural stabilisation concept with „dead-zone“.

outputs and ANN structures size (16-8-4-1) were retained. The ANN input vectors consisted of 20 samples of the respective phase currents, 10 from the HV and 10 from the MV side, taken at every second time instant within full cycle data window (sampling at 1kHz).

The protection with stabilisation unit as shown in Fig. 2 proved to be reliable for all transformer energisation cases (inrush confirmed) and high-current fault cases (tripping allowed). With the threshold set at 0.0 the protection had problems with classifying some internal winding fault cases for which current values differed very little from the ones observed under nominal operating conditions (non-fault).

Further improvement of the scheme that led to the final stable and reliable protection version was connected with introduction of the „dead-zone“, according to the formula:

$$blocking = \begin{cases} 1 & \text{if } F_{av} \geq D \\ 1 & \text{if } -D < F_{av} < D \wedge S_{2h} = 1 \\ 0 & \text{if } -D < F_{av} < D \wedge S_{2h} = 0 \\ 0 & \text{if } F_{av} \leq -D \end{cases} \quad (1)$$

where: F_{av} – mean value of the ANN outputs, D – „dead-zone“ width, S_{2h} – blocking signal from the 2nd harmonic restraint.

When the absolute value of ANN output F_{av} is low, the stabilisation task is left to 2nd harmonic criterion (illustration in Fig. 3). Big advantage of such an approach is an opportunity of adjusting the uncertainty region of any width. Simulative testing of the protection has shown that setting „dead-zone“ width D at the level 0.1 was enough to make scheme perform-

ance errorless, even for low-current internal winding faults. It has been found that increasing the value of D (widening the „dead-zone“) may on the one hand help to stabilise the relay output in the transient period after disturbance inception but on the other hand may have unfavourable impact on the speed of decision making. The slowest operation of such a double-criterion stabilisation is limited by the characteristics of the 2nd harmonic blocking when D is set at maximum, i.e. at 1.0.

III. TESTING OF THE SCHEMES

The developed neural stabilization schemes described in previous Section have been tested in simulative way for the model of an HV/MV 32MVA 115/22kV YNd11 five-leg core transformer that is depicted Fig. 4. The protected transformer was a part of wider system model consisting also of equivalent feeding systems from both sides (represented by electromotive sources behind appropriate impedances, $X_0/X_1=1.5$ or 1.0 for HV and MV system, respectively). The protected transformer could be fed from the HV side, MV side and from both sides. Additional load might be connected on either of the transformer sides (opposite as seen from the supplying side).

To obtain a wide variety of transformer fault and inrush cases the following parameters of the system and transient conditions were being changed in each simulation run:

- transformer feeding (from the HV side, MV side, or both),
- strength of the supplying sources (pu system impedance: 0.02, 0.1, 0.3 or 0.5),
- star point operation mode (grounded, isolated),
- fault point:
 - at selected transformer side (HV or MV),
 - external or internal (out-zone or in-zone),
 - terminal or internal winding short-circuit,
- transformer energisation
 - without load or on-load,
 - from the HV or MV side,
 - at various time instants,
 - for two values of remanent flux (0, max),
- mixed cases (energisation followed by internal fault, etc.).

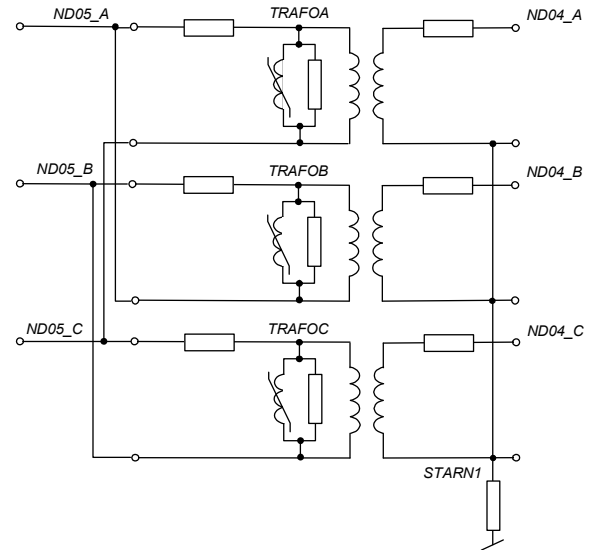


Fig. 4. Schematic diagram of the HV/MV transformer under study.

TABLE I. BREAKDOWN OF THE PERFORMANCE PARAMETERS OF THE ANN-BASED RELAYS STUDIED.

Type of transient	Tripping time [ms]			
	Standard protection	Protection with ANN stabilization		
		<i>A</i>	<i>B</i>	<i>C</i>
Energisation cases	–	–	–	–
External faults	–	–	–	–
Internal fault (winding – 8%)	21	11	11	11
Internal fault (winding – 15%)	19	3	7	3
Internal fault (winding – 80%)	21	3	3	2
Internal fault (L1–L3, HV side)	145	2	2	2
Internal fault (L2–G, HV side)	21	3	3	2
Internal fault (3-ph, HV side)	267	2	2	2
Internal fault (L1–L3, MV side)	7	3	3	2
Internal fault (L2–G, MV side)	21	11	11	15
Internal fault (3-ph, MV side)	71	3	5	2

With the abovementioned options a few hundred of simulation cases have been generated. Additional real-world signals originating from fault recorders were also used for testing of the developed protection schemes. In Table I the results of ANN-stabilized protection testing are gathered for all versions of relay arrangement as described above. The scheme testing was done with 70 selected EMTP-ATP simulation cases presenting various conditions of transformer operation. The tripping times shown in the table are mean values for all cases of given class. It is worth to mention that in cases of high current faults the overcurrent support (condition 4, Fig. 1) could be superior to other decision criteria and evoke fast tripping. From Table I it is seen that both traditional and intelligent protection versions studied were stable for all cases of external faults, i.e. no overfunction was observed [11].

Unquestionable advantage of the developed ANN-based protection schemes is much faster response than the traditional non-AI relay version, especially for high current faults occurring at the HV side of the protected transformer. Speeding-up of the relay was often not only an effect of ANN-based stabilisation but also a result of overcurrent support whose reaction could overlap or sometimes even pass the response of the ANNs applied. For many cases, however, when the overcurrent criterion was not activated the relay response was resulting only from the decision speed of the neural stabilisation.

The AI protections developed were using one or three neural networks for stabilisation purpose. A shortcoming of the single-ANN unit was a possibility of lack of stabilisation and erroneous transformer tripping for some inrush cases that were classified as internal faults. Much better reliability was reached for three-ANN solutions that were characterised by perfect selectivity of discrimination between magnetising inrush and

in-zone fault cases. The protection solution whose stabilisation unit was fed with transformer terminal currents from both sides could recognise the majority of the internal faults faster than the solution fed with differential signals. Additional virtue of the protection version with “dead-zone” was the usage of both ANN and 2nd harmonic stabilisation (adaptivity with relation to the ANN output values), which favourably affected dependability and speed of the relay stabilisation for inrush cases.

IV. ANN STRUCTURE OPTIMIZATION

Discussed above results of ANN-based stabilization units were obtained for ANNs selected with the heuristic “trial and error” method. Since one cannot claim that such an approach to ANN design is effective and brings optimal decision units, further research has been done in order to find the most useful ANN structures in a more ordered manner, i.e. with genetic algorithm. In Fig. 5 the adopted block scheme of the genetic optimization procedure is presented [12]. At the beginning a so-called initial population of neural networks is randomly created. The ANNs are trained with selected typical patterns and validated with all available patterns. After the quality of each individual was determined (quality index calculated), an intermediate population is created where successful individuals are reproduced more likely. On this intermediate population several different genetic operations can be applied to create a new population. The most important one is the crossover of two parent individuals to produce new descendants. Furthermore, mutations may take place, which change the network topology randomly by adding or removing neurons. Mutations are used to avoid that the optimization is done around a local minimum. The consecutive populations of ANNs are created, trained and graded in a closed loop until the selection criterion is fulfilled or the prescribed number of generation is reached. The evolutionary process described should end up in an optimum that represents the most appropriate network topology.

The GA scheme has successfully been used by the authors in ANN optimization for CT saturation detection module [12]. Here, the procedure was applied for optimization of proposed ANN-based stabilisation of transformer differential protection. The results given below are related to the single-ANN stabilisation scheme (*A*), however, similar processing was done also for the ANNs of the schemes *B* and *C*. Depending on the definition of the quality index Q adopted for ANN grading, the following optimal neural networks were obtained:

- (33-20-1), for $Q_1=1-mse$,
- (18-1), for $Q_2=(1-mse)*eff$,
- (12-1), for $Q_3=(1-mse)*eff/size$,

whereas: mse – mean squared error of ANN output, eff – efficiency of stabilisation, $size$ – total number of ANN neurons.

One can see that except of the network optimized with the aim to minimize the mse value (maximize the quality index Q_1), much more compact ANN structures are gained with application of the genetic procedure for the other quality measures. Tests confirmed that despite small ANN size (less than 20 neurons) the structures obtained assure successful inrush recognition for all simulated inrush cases.

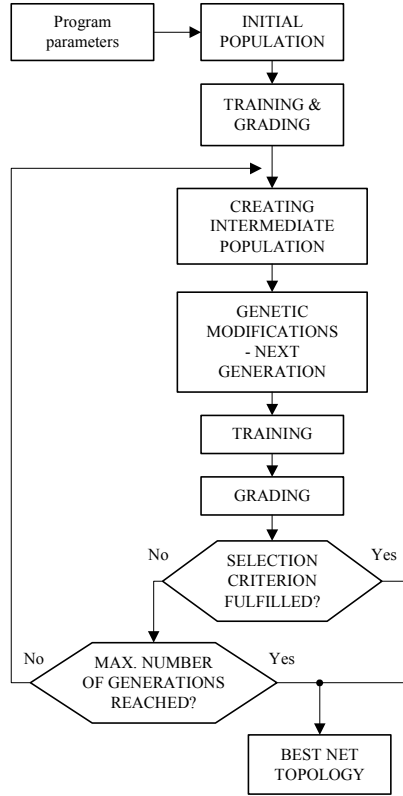


Fig. 5. Flow chart of the GA procedure for the ANN topology optimization.

V. ANN-STABILIZATION PERFORMANCE

The designed neural protection schemes were tested with both EMTP-ATP generated signals and real-world registered cases. The performance of proposed ANN-based stabilization units is compared with 2nd harmonic based restraint scheme.

In Fig. 6 the differential current (for chosen phase), second harmonic content and the decision signals are shown for a case of severe double-phase in-zone fault at the HV terminals of protected transformer is shown. One can see that the second harmonic blocking remained active over several cycles thus preventing the relay from expected prompt tripping. All the designed neural schemes reacted properly with tripping decision issued within maximum 5 ms after fault inception.

In Fig. 7 a case of transformer energization (unloaded) with CT saturation is shown. The 2nd harmonic content was high enough to assure blocking of the traditional relay. All the proposed neural schemes also responded with expected blocking, no maloperation was observed.

Interesting case of transformer energization being a record from real-world can be studied in Fig. 8. The traditional stabilization was ineffective here because of quite low content of the 2nd harmonic signal, which led to undesired tripping of the transformer some 50 ms after the decision was issued (CB time). Among the three versions of ANN-based protection schemes only the one fed with transformer terminal currents and “dead-zone” was fully stable in this case. The output signals of the other ANN schemes were not reliable, with the single-ANN scheme showing by far the worst performance.

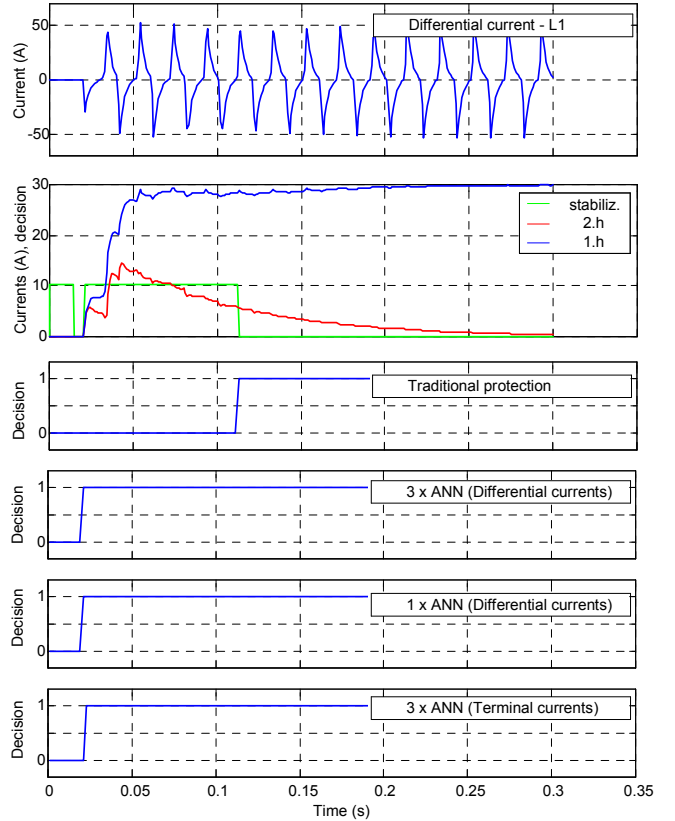


Fig. 6. Differential, stabilization and decision signals for the case of A-C terminal fault at the HV side.

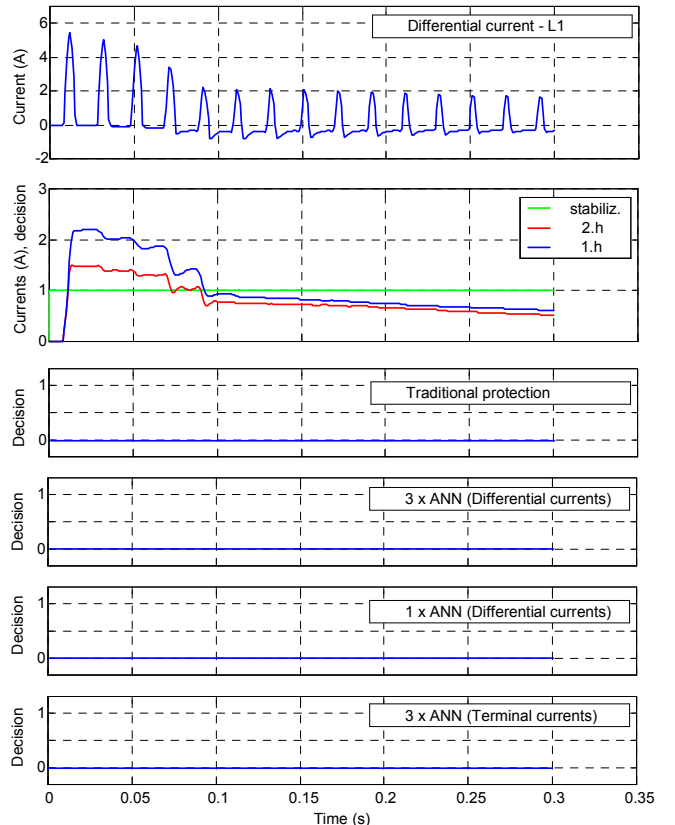


Fig. 7. Differential, stabilization and decision signals for the case of energizing of the unloaded transformer.

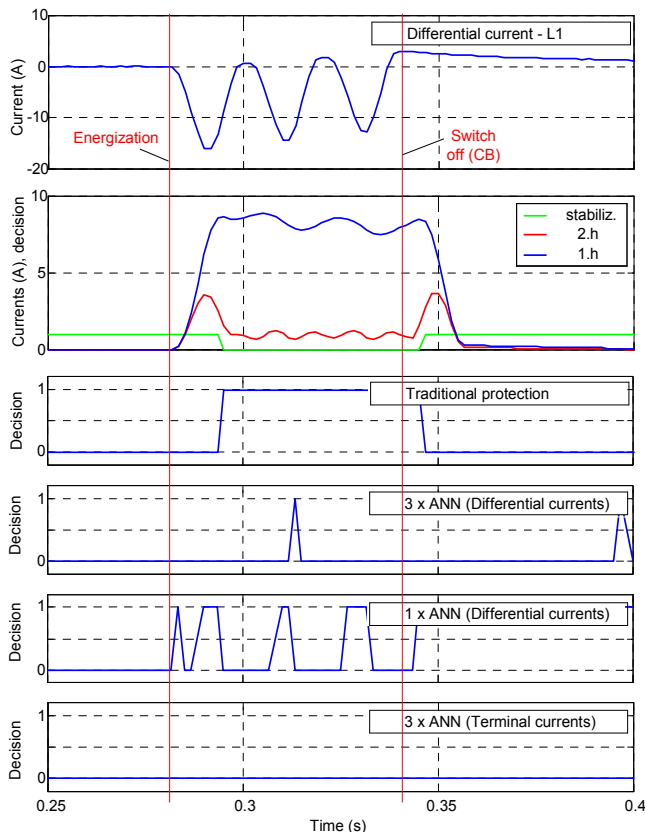


Fig. 8. Differential, stabilization and decision signals for the case of transformer energization with false relay response (real-world recorded case).

VI. CONCLUSIONS

New ANN-based stabilization schemes for transformer differential protection have been developed and described in the paper. Out of the three versions proposed the one with three neural networks and dead-zone, fed with transformer terminal currents, brought about the best results. All the schemes proved to be a very good tool for detecting magnetising inrush.

Application of the proposed neural stabilization units allows for faster excluding of magnetising inrush condition during faults, which can be also seen as a mean of relay response speed-up. Nevertheless, further testing of the scheme with real-world signals is needed to assure faultless performance under any possible operating conditions.

Possibility of genetic algorithm application for optimization of ANN based inrush detection units has also been studied. The resulting ANNs are characterized by compact structures with maintained recognition abilities as for initial much bigger networks. Laboratory hardware tests confirmed that real-time implementation of such small ANNs is utterly attainable even with use of moderate speed signal processors, with ANN execution time being a fraction of a millisecond [13].

VII. REFERENCES

- [1] Neibur D. (Convenor), *Artificial neural networks for power systems*, Report by TF 38.06.06, Electra No. 159, April 1995.
- [2] Dillon T.S. (Convenor), *Fault diagnosis in electric power systems through AI techniques*, Report by TF 38.06.02, Electra, No. 159, April 1995.

- [3] Saha M.M., Kasztenny B., *AI methods in power system protection*, Eng. Int. Systems, Vol. 5, No. 4, Dec. 1997, s. 183–184.
- [4] Kezunovic M., et. al., *Practical intelligent system applications to protection, and substation monitoring and control*, Proceedings of the 1998 CIGRE Session, Paris, France, 1998, Paper 34–104.
- [5] Horowitz S.H., Phadke A.G., *Power system relaying*, Wiley & Sons, New York, 1992.
- [6] Kasztenny B., Rosolowski E., Saha M.M., Hillstrom B., *A comparative analysis of protection principles for multi-criteria power transformer relaying*, Proceedings of the 12th PSCC Conference, Dresden, Germany, 1996, pp. 107–113.
- [7] Sidhu T.S., Sachdev M.S., Wood H.C., *Detecting transformer winding faults using nonlinear models of transformers*, Proc. of the 4th Int. Conference DPSP, IEE Publ. No. 302, 1989, pp. 70–74.
- [8] Thorp J.S., Phadke A.G., *A new computer based flux restrained current differential relay for power transformer protection*, IEEE Trans. on PAS, Vol. PAS-102, No. 11, Nov. 1983, pp. 3624–3629.
- [9] Kasztenny B., Rosolowski E., Saha M.M., Hillstrom B., *A self-organizing fuzzy logic based protective relay – an application to power transformer protection*, IEEE Transactions on Power Delivery, Vol. 12, No. 3, July 1997.
- [10] *Numerical differential protection relay for transformers, generators, motors and mini busbars*. 7UT613/63x V.4.06 Instruction Manual, Order No. C53000-G1176-C160-2. SIEMENS AG 2006.
- [11] Palsy W., *Stabilisation of transformer differential relays against magnetising inrush* (in Polish), Master Thesis, Wroclaw University of Technology, Poland, 2005.
- [12] Rebizant W., Bejmert D., *Current Transformer Saturation Detection with Genetically Optimized Neural Networks*, Proceedings of the 2005 IEEE PowerTech Conference, St. Petersburg, Russia, 27–30 June 2005, CD-ROM, paper 220.
- [13] Rebizant W., Bejmert D., Staszewski J., Schiel L., *CT Saturation Detection and Correction with Artificial Neural Networks*, 2nd International Conference on Advanced Power System Automation and Protection, APAP2007, Jeju, Korea, April 24–27, 2007, paper 504.

VIII. BIOGRAPHIES



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Daniel Bejmert was born in 1979 in Walbrzych, Poland. He graduated from the Electrical Engineering Faculty of Wroclaw University of Technology, Poland in 2004. At present he is a PhD student at the Institute of Electrical Power Engineering of WUT. His research interests include application of intelligent algorithms in digital protection and control systems.



Ludwig Schiel was born in Weimar, Germany, in 1957. He studied Electrical Engineering at the Institute of Technology Zittau, Germany, finishing with the Dipl.-Ing. degree in 1984. In 1991 he received the Dr.-Ing. degree. In the same year he joined the Siemens AG, Germany, Department of Power Transmission and Distribution, Energy Automation. He is project manager of transformer differential protection systems.